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import pandas as pd
import matplotlib.pyplot as plt
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rulesw

# Step 1: Sample transaction data
transactions = [
    ['Milk', 'Bread', 'Butter'],
    ['Bread', 'Butter'],
    ['Milk', 'Bread'],
    ['Milk', 'Butter'],
    ['Bread', 'Butter'],
    ['Milk', 'Bread', 'Butter']
]

# Step 2: One-hot encoding
te = TransactionEncoder()
te_data = te.fit(transactions).transform(transactions)
df = pd.DataFrame(te_data, columns=te.columns_)

# Step 3: Apriori
frequent_items = apriori(df, min_support=0.3, use_colnames=True)

# Step 4: Association rules
rules = association_rules(frequent_items, metric="confidence",
min_threshold=0.6)

# Print results
print("Frequent Itemsets:\n", frequent_items)
print("\nAssociation Rules:\n",
rules[['antecedents','consequents','support','confidence','lift']])

# Step 5: Plot - Support of itemsets
plt.figure()
plt.bar(range(len(frequent_items)), frequent_items['support'])
plt.xticks(range(len(frequent_items)), frequent_items['itemsets'],
rotation=45)
plt.title("Frequent Itemsets Support")
plt.xlabel("Itemsets")
plt.ylabel("Support")
plt.tight_layout()
plt.show()

# Step 6: Scatter Plot (Support vs Confidence)
plt.figure()
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plt.scatter(rules['support'], rules['confidence'])
plt.title("Support vs Confidence")
plt.xlabel("Support")
plt.ylabel("Confidence")
plt.show()
```

```
Encoded Dataset:

   Bread  Butter   Milk
0   True    True   True
1   True    True  False
2   True   False   True
3  False    True   True
4   True    True  False

Frequent Itemsets:

   support         itemsets
0      0.8          (Bread)
1      0.8         (Butter)
2      0.6           (Milk)
3      0.6  (Bread, Butter)
4      0.4    (Bread, Milk)
5      0.4   (Milk, Butter)

Association Rules:

  antecedents consequents  support  confidence      lift
0     (Bread)    (Butter)      0.6    0.750000  0.937500
1    (Butter)     (Bread)      0.6    0.750000  0.937500
2      (Milk)     (Bread)      0.4    0.666667  0.833333
3      (Milk)    (Butter)      0.4    0.666667  0.833333
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: Deprecat
  return datetime.utcnow().replace(tzinfo=utc)
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